

CURRICULUM VITAE

AKSHAY KARGAONKAR

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SUMMARY

Mechanical Engineer with hands-on experience in designing and scaling reliable, cost-effective hardware for mass production. Skilled in CAD, simulation, and rapid prototyping with a strong understanding of integrating embedded electronics with actuators and power transmission systems.

Proven experience managing the complete product lifecycle from zero-to-one development to establishing ISO 9001 Quality Management Systems and coordinating global manufacturing operations. Red Dot Design Award winner for sustainable engineering solutions.

WORK EXPERIENCE

Development and Production Manager

Mar 2026 - Present

DG TAKANO Co., Ltd. ([website](#))

Osaka, Japan

- **R&D Management:** Leading the R&D team in the development of innovative sanitary and kitchenware products aligned with the company's water-saving goals.
- **Production Management:** Managing in-house production operations at the Higashiosaka manufacturing facility, including production scheduling, workflow coordination, inventory management, and ERP-based production control.
- **Global Manufacturing & Supply Chain:** Connected DG TAKANO with manufacturing partners across China, South Korea, and Vietnam, reducing production lead time and manufacturing costs by 40%.
- **Cross-Functional Collaboration:** Collaborated with international industrial design firms including California based Whipsaw Inc. to align engineering feasibility with premium product design standards.

Research and Development Engineer

Sept 2022 - Mar 2026

DG TAKANO Co., Ltd. ([website](#))

Osaka, Japan

- **Product Development:** Led zero-to-one product development and sales-strategy planning for the [5a-faucet](#), a Red Dot Award-winning water-saving faucet integrating high pulsation flow. Now sold across 200 Yamada Denki stores around Japan.
- **Mechatronics Design & Testing:** Designed and developed a faucet head with an autonomous switching mechanism between laminar and aerated flows, improving water efficiency by up to 60% based on user interaction and sensor input.
- **Sensor Integration & Validation:** Developed a government-subsidized [smart kitchen platform](#) by embedding flow and temperature sensors into a custom-designed faucet body and supporting AWS-based cloud connectivity for remote data collection and analysis.
- **Durability Engineering:** Engineered physical endurance-testing rigs and conducted extensive durability testing on multi-axis faucet joints, simulating over 100,000 motion cycles across multiple degrees of freedom.
- **Prototyping & Fabrication Tools:** Programmed and operated Nakamura-Tome multi-turret lathe machines using G-code and performed rapid prototyping with KEYENCE Polyjet 3D printers, Autodesk Fusion 360, SolidWorks, and MIRO.

EDUCATION

Indian Institute of Technology, Dhanbad, India (0.5% Acceptance rate)

2018 - 2022

Bachelor of Technology in Mechanical Engineering CGPA: 8.01/10.00 ([degree](#))

Relevant Coursework- Kinematics of Machines, Dynamics of Machinery, Advanced Solid Mechanics, Finite Element Analysis, Rapid Prototyping, Robotics ([Transcripts](#))

RESEARCH LAB EXPERIENCE

Formula Student - Mechismu Racing, IIT ISM

Design Head

2019 - 2022

- Engaged in advanced **modeling and simulation of the suspension system** for the IIT ISM Formula Student vehicle using MSC ADAMS, contributing to enhanced stability of the vehicle. [\(link\)](#)
- Directed a thorough analysis of **driver ergonomics to reduce injuries** from steering torque, showcasing a keen focus on performance enhancement while ensuring safety.

Laboratorio de Ingeniería Mecánica, UDC Spain

Project under Prof. Javier Cuadrado

Oct 2020 - Apr 2021

Remote, India

- Collaborated with Professor Cuadrado’s team to conduct **real-time co-simulation** of multiphysics systems, fostering interdisciplinary insights and advancements in system dynamics and integration.
- Utilized MATLAB to create and simulate benchmark mechanical subsystems, employing various solving methods and **streamlining simulation time** for optimal performance evaluation.

PUBLICATIONS

Akshay Kargaonkar, N.K. Singh, “**Multibody Dynamic Simulation of A Mountain Bike To Reduce Riding Injuries During A Jump Event,**” in *IJERT Journal*, vol. 12, issue 6, 29 June, 2023. [DOI](#)

Akshay Kargaonkar, N.K. Singh, “**A Study on Modeling and Simulation of different Friction Models in Multibody Dynamics,**” in *EJMS Journal*. (submitted) [\(link\)](#)

SKILLS

CAD softwares	Autodesk Fusion360, CATIA V5, SolidWorks
Simulation softwares	ANSYS Mechanical, FLUENT, ADAMS, COMSOL Multiphysics
Programming languages	C, C++, Python, MATLAB
Languages	English(Bilingual proficiency), Japanese(Limited working proficiency)
Others	Worked with various Sensors, Micro-controllers and Development boards

ACHIEVEMENTS

- **Red Dot: Best of the Best award 2025** for the 5a faucet as a Product Engineer. [\(link\)](#)
- Contributed to the design and development of DG Takano’s MeliorDesign Dishware, which received the **Red Dot Design Award: Product Design 2024**. [\(link\)](#)
- **IELTS Academic** overall score: **8.5**, R: 9.0, L: 9.0, W: 7.5, S: 7.5 - November, 2023.
- **JLPT N4** - July, 2025.